Kai-Chieh Hsu

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I build safe physical AI for autonomous systems, combining world–action models, vision–language–action models, and flow-based generative models.

Experience

Senior Machine Learning Research Scientist
Mountain View, CA

Nuro Inc. (Manager: Li Zhou)

Senior Machine Learning Research Scientist — June 2026 - Present

Machine Learning Research Scientist — July 2024 - May 2026

- Develop foundation models for autonomous driving, including vision–language–action models and world–action models, unifying planning, perception, and prediction end-to-end.
- Fine-tune flow-based generative models for action generation, improving safety compliance and long-tail generalization.

Research Assistant
Princeton, NJ

Princeton University, Safe Robotics Laboratory (Advisor: Jaime Fernández Fisac) — Aug. 2020 - May 2024

My work applies algorithmic and theoretical techniques from machine learning, control theory, and game theory to ensure the safety of learning-enabled autonomous systems. These algorithms scale to high-order dynamic systems and can handle complex deployment conditions. Additionally, they systematically unify the autonomy stack to prevent unwarranted conservativeness.

Engineering Intern
San Diego, CA

Qualcomm Research (Manager: Stephen Chaves, Mentor: Pranav Desai) — May 2023 - Aug. 2023

I Proposed a **unified neural backbone** for agent predictor and behavior planner in autonomous vehicles software stack and used reinforcement learning and imitation learning for implementing behavior planners.

Research Scientist Intern
Remote

NVIDIA Research, Autonomous Vehicle (Manager: Prof. Marco Pavone, Mentor: Prof. Karen Leung, Yuxiao Chen) — May 2022 - Dec. 2022

I Formalized **responsibility** by safety margin decrease and policy shift with **counterfactual reasoning** and estimated the responsibility level online with **hidden Markov model**. I incorporated the estimated responsibility into **trajectory prediction** models to improve prediction accuracy and interpretability.

Education

Princeton University
Princeton, NJ

Ph.D. in Electrical and Computer Engineering — Sept. 2021 - May 2024

M.A. in Electrical and Computer Engineering — Sept. 2019 - May 2021

- Concentration: Machine learning and Robotics
- 4.0/4.0 GPA
- Advisor: Prof. Jaime Fernández Fisac
- Thesis: Scaling Full-Stack Safety for Learning-Enabled Robot Autonomy

National Taiwan University
Taipei, Taiwan

B.S. in Electrical Engineering — Sept. 2014 - Jan. 2019

- Concentration: Signal processing and Digital IC design
- 4.19/4.30 GPA and ranked in **top 10** of the class
- Research Advisors: Prof. An-Yeu (Andy) Wu and Prof. Jean-Fu Kiang

Selected Publications

Journal Papers

[J1] **K.-C. Hsu**, Haimin Hu, and J. F. Fisac, The Safety Filter: A Unified View of Safety-Critical Control in Autonomous Systems, in *Annual Review of Control, Robotics, and Autonomous Systems,* Feb 2024.

[J2] A. R. Kumar, **K.-C. Hsu**, P. J. Ramadge, and J. F. Fisac, Fast, Smooth, and Safe: Implicit Control Barrier Functions through Reach-Avoid Differential Dynamic Programming, in *IEEE Control Systems Letters,* vol. 7, pp. 2994-2999, June 2023.

[J3] **K.-C. Hsu***, A. Z. Ren*, D. P. Nguyen, A. Majumdar+, and J. F. Fisac+, Sim-to-Lab-to-Real: Safe Reinforcement Learning with Shielding and Generalization Guarantees, in *Artificial Intelligence,* Jan 2023. | Spotlight in ICLR Workshop and NeurIPS Workshop

[J4] C.-Y. Chou, **K.-C. Hsu**, B.-H. Cho, K.-C. Chen, and A.-Y. (Andy) Wu, Low-Complexity On-demand Reconstruction for Compressively Sensed Problematic Signals, in *IEEE Transactions Signal Processing,* vol. 68, pp. 4094-4107, July 2020.

Conference Papers

[C1] D. P. Nguyen*, **K.-C. Hsu***, Wenhao Yu, Jie Tan, and J. F. Fisac, Gameplay Filters: Robust Zero-Shot Safety through Adversarial Imagination, in *Conf. Robot Learning,* Munich, Germany, Nov 2024.

[C2] H. Hu, K. Nakamura, **K.-C. Hsu**, N. E. Leonard, and J. F. Fisac, Emergent Coordination through Game-Induced Nonlinear Opinion Dynamics, in *Proc. IEEE Conf. Decision and Control,* Singapore, Dec 2023.

[C3] **K.-C. Hsu**, K. Leung, Y. Chen, J. F. Fisac, and M. Pavone, Interpretable Trajectory Prediction for Autonomous Vehicles via Counterfactual Responsibility, in *IEEE/RSJ Int. Conf. Intelligent Robots & Systems,* Detroit, MI, USA, Oct 2023.

[C4] **K.-C. Hsu***, D. P. Nguyen*, and J. F. Fisac, ISAACS: Iterative Soft Adversarial Actor-Critic for Safety, in *Learning for Dynamics & Control,* Philadelphia, PA, USA, Jun 2023.

[C5] **K.-C. Hsu***, V. Rubies-Royo*, C. J. Tomlin, and J. F. Fisac, Safety and Liveness Guarantees through Reach-Avoid Reinforcement Learning, in *Proc. Robotics: Science and Systems,* Virtually, July 2021.

Awards and Honors

Hon Hai Technology Award — Hon Hai Education Foundation, Taiwan, June 2024
Rewarding outstanding performance in robotics

Bede Liu Fund for Excellence — Princeton University, NJ, Oct. 2023, Mar. 2024

Travel Grant — Princeton University, NJ, Nov. 2022
From the School of Engineering and Applied Science

Teaching Assistant Award — Princeton University, NJ, Sept. 2022
For the new *Intelligent Robotic Systems* course

3rd Prize in Integrated Circuit Design Contest — Ministry of Education, Taiwan, July 2018
Out of about 300 teams

2nd Prize in Taiwan Creative Electromagnetic Implementation Competition — High-speed RF and mm-Wave Tech. Center, Taiwan, Aug. 2017
Under the supervision of Prof. Tzong-Lin Wu |

8th place in Cadence Data Structure and Programming Contest — Taipei, Taiwan, Mar. 2017
Out of about 250 students

Graduate Representative — National Taiwan University, Taiwan, June 2018
Top ten students in the Department of Electrical Engineering

Professor Chun-Hsiung Chen Scholarship — Electromagnetic Industry-Academia Consortium, Taiwan, Jan. 2018
Rewarding outstanding performance in electromagnetic research

Dean's List — National Taiwan University, Taiwan, June 2014, June 2016